Class 9

1 \ 2	L	R
B	(2,	(1,
T	(4,	(0,

(1.5,	(1
(3.5,	(0

$(p_1, p_2, p_3, 1-p_1-p_2-p_3)$ $\quad p_i \geq 0$

$(B,L), (B,R), (T,L), (T,R)$

$\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$

Embedded matrix

$1/4 \times 2 + \frac{1}{4} \times 1 + \frac{1}{4} \times 4 + \frac{1}{4} \times 0$

$= \frac{7}{4}$

	L	R
B	$(2\alpha+\beta$	$(\alpha+\beta$
T	$(4\alpha+\beta$	$(\beta$

$\alpha > 0$

Example

	L	R	
T	4	1	1
B	2	3	2

$\underline{v}_1 = 2$

	L	R
1/4 T	4	1
3/4 B	2	3

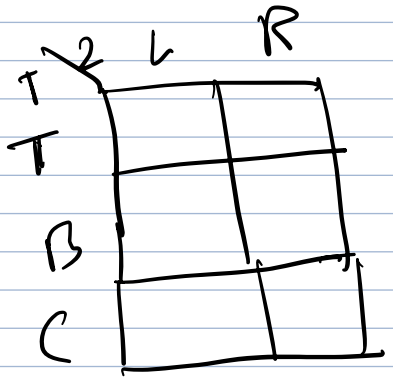
$$\min_{s_2 \in \{L,R\}} u\left(\tfrac{1}{4}[T], \tfrac{3}{4}[B], s_2\right)$$

$$= \min\left\{4 \times \tfrac{1}{4} + 2 \times \tfrac{3}{4},\ 1 \times \tfrac{1}{4} + 3 \times \tfrac{3}{4}\right\}$$

$$= \min\left\{\frac{10}{4}, \frac{10}{4}\right\}$$

$$= \frac{10}{4}$$

Defn Let $G = \langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$ be a strategic form game in which the set of strategies of all the players is finite. A mixed strategy of Player i is a prob distribution over S_i. The set of all mixed strategies of Player i is denoted by $\Sigma_i := \left\{\sigma_i : S_i \to [0,1] \;\middle|\; \sum_{s_i \in S_i} \sigma_i(s_i) = 1\right\}$



$\sigma_1 = [p_1(T), 1-p_1(B)]$

$\sigma_1' = [p_1(T), p_2(B), 1-p_1-p_2(C)]$

Mixed extension of a game

Defn Let $G = \langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$ be a strategic form game where all the players have finitely many strategies. Then the mixed extension of the game $\Gamma = \langle N, (\Sigma_i)_{i\in N}, (V_i)_{i\in N}\rangle$

- Σ_i the set of mixed strategies of Player i
- $V_i(\sigma_1, \sigma_2, \ldots, \sigma_n) = \sum_{(s_1, s_2, \ldots, s_n) \in S_1 \times S_2 \times \cdots \times S_n} u_i(s_1, \ldots, s_n)\sigma_1(s_1)\sigma_2(s_2)\cdots\sigma_n(s_n)$

Defn Suppose G_1 is a game in strategic form and Γ is a mixed extension of G_1. Then Every NE of Γ is called a NE in mixed strategies of G_1. Suppose G_1 is a two players zero sum game and Γ has a value v then we say G_1 has value v in mixed strategies.

- A profile $\sigma^* = (\sigma_1^*, \sigma_2^* \dots, \sigma_n^*)$ is a NE in mixed strategies of a game G_1 if

$$U_i(\sigma^*) \geq U_i(\sigma_i, \sigma_{-i}^*) \quad \forall i \in N, \ \forall \sigma_i \in \Sigma_i$$

$$\Rightarrow U_i(\sigma^*) \geq U_i(s_i, \sigma_{-i}^*) \quad \forall i \in N, \ \forall s_i \in S_i$$

$\Leftarrow$

Suppose $U_i(\sigma^*) \geq U_i(s_i, \sigma_{-i}^*) \quad \forall i \in N, \ \forall s_i \in S_i$

Consider player i and a mixed strategy σ_i

$$U_i(\sigma_i, \sigma_{-i}^*) = \sum_{s_i \in S_i} U_i(s_i, \sigma_{-i}^*)\, \sigma_i(s_i) = \sum_{s_j \in S_j} U_i(s_j, \sigma_{-j}^*)\, \sigma_j(s_j)$$

$$\leq \sum_{s_i \in S_i} U_i(\sigma^*)\, \sigma_i(s_i)$$

$$= U_i(\sigma^*) \sum_{s_i \in S_i} \sigma_i(s_i) = U_i(\sigma^*)$$

- $U_i(\sigma_1, \sigma_2, \ldots, \lambda\sigma_j + (1-\lambda)\sigma_j', \sigma_{j+1}, \ldots, \sigma_n)$

$$= \lambda U_i(\sigma_1, \sigma_2, \ldots, \sigma_j, \sigma_{j+1}, \ldots, \sigma_n) + (1-\lambda) U_i(\sigma_1, \sigma_2, \ldots, \sigma_j', \sigma_{j+1}, \ldots, \sigma_n)$$